
DevOps Scripting Roadmap (CTA Resource)

Objective

Learn the **minimum effective scripting skills** required to succeed as a DevOps Engineer using:

- Bash
- Python
- YAML

1. Bash Scripting (Foundation First)

◆ Why Bash?

Bash is the backbone of Linux-based systems where most DevOps work happens.

◆ What to Learn

✓ Core Commands

- File operations: `ls`, `cp`, `mv`, `rm`
- Text processing: `grep`, `awk`, `sed`
- Permissions: `chmod`, `chown`

✓ Scripting Basics

- Variables and environment variables
- Conditional statements (`if`, `case`)
- Loops (`for`, `while`)
- Functions

✓ Automation Use Cases

- Server setup scripts
- Log cleanup scripts
- Backup automation

- Cron jobs scheduling

✓ Example

```
#!/bin/bash
echo "Starting deployment..."
git pull origin main
npm install
pm2 restart app
echo "Deployment completed!"
```

2. Python for DevOps (Automation Power)

◆ Why Python?

When Bash becomes complex, Python simplifies logic and integrates easily with APIs and cloud tools.

◆ What to Learn

✓ Core Concepts

- Variables, data types
- Loops and conditions
- Functions and modules

✓ DevOps-Focused Topics

- File handling (read/write logs, configs)
- API calls (`requests` library)
- JSON & YAML parsing
- Error handling

✓ Libraries to Focus

- `os, sys` → system operations
- `subprocess` → run shell commands
- `requests` → API automation
- `boto3` → AWS automation

✓ Use Cases

- Automating cloud resource creation
- Monitoring scripts
- CI/CD pipeline integrations
- Log analysis

✓ Example

```
import os

print("Deploying application...")
os.system("git pull origin main")
os.system("docker-compose up -d")
print("Deployment done!")
```

3. YAML (Configuration Language – Must Have)

◆ Why YAML?

Used to define pipelines, infrastructure, and Kubernetes resources.

◆ Where You'll Use It

- CI/CD pipelines (GitHub Actions, Azure DevOps)
- Kubernetes manifests
- Docker Compose files

◆ What to Learn

✓ Basics

- Key-value pairs
- Lists and indentation
- Comments

✓ Intermediate

- Variables and parameters
- Anchors & aliases
- Multi-stage pipelines

✓ Example (CI/CD)

```
trigger:
  - main

jobs:
  - job: Build
    steps:
      - script: echo "Building app"
```



How These 3 Work Together

Tool	Role in DevOps
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Bash	Quick automation & system tasks
Python	Complex automation & integrations
YAML	Define pipelines & infrastructure

👉 Example Flow:

- YAML defines pipeline
 - Bash runs commands
 - Python handles logic/API calls
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Recommended Learning Path

Week 1–2

- Linux basics + Bash scripting

Week 3–4

- Python fundamentals + automation scripts

Week 5

- YAML + CI/CD pipelines

Week 6+

- Combine all in real projects